AI Playbook and Toolkit for Public Health Departments

AI Documentation, Model Cards, and System Cards

GOV 320

AI documentation makes systems understandable, reviewable, and accountable. Model cards and system cards are structured documentation tools that describe intended use, data, evaluation, limitations, risks, human oversight, and monitoring. This module helps public health staff create documentation that supports governance review, vendor oversight, public accountability, and incident response.

Level	intermediate/practitioner
Primary track	Governance and Security Track
Appears in tracks	operations-data-quality

Learning Objectives

- Explain the purpose of ai documentation, model cards, and system cards in public health AI work.
- Identify the technical, operational, privacy, security, equity, and governance issues associated with the topic.
- Apply the topic to a real or realistic public health workflow, system, or implementation scenario.
- Recognize common failure modes that can affect safety, accuracy, trust, workflow fit, or sustainability.
- Identify practical guardrails that should be documented before implementation.
- Produce a AI documentation packet that can support readiness assessment, governance review, procurement, implementation, or monitoring.

Module Overview

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Jurisdiction and Agency Policy Note

This national-level module is intended for training and planning purposes. It does not replace legal, privacy, procurement, civil rights, records, or agency policy review. Learners should confirm applicable requirements in their own jurisdiction and organization before implementing AI-supported workflows.

Why This Topic Matters for Public Health AI

AI Documentation, Model Cards, and System Cards matters because AI systems are embedded in public health programs, data systems, and decision processes rather than operating in isolation. The topic affects whether AI outputs are accurate, explainable, safe, equitable, secure, and sustainable. Agencies that ignore this area may create tools that work in a demonstration but fail in actual practice.

The topic also affects accountability. Public health agencies must be able to explain how data are used, why a tool was approved, what evidence supports the workflow, who reviews outputs, and what happens when the system fails. These responsibilities become more important as AI tools move from experimentation into operational use.

Core Concepts

AI documentation records what the system is intended to do, how it was developed or configured, what data it uses, how it was evaluated, what limitations are known, and how it is monitored. Documentation supports accountability across the AI lifecycle.

A model card is a structured summary of a model, including intended use, data, evaluation, limitations, risks, and appropriate use. A system card extends documentation to the broader system, including workflow integration, interfaces, user roles, retrieval sources, human oversight, and operational controls.

Documentation is not a paperwork exercise. It is how agencies preserve institutional memory, support governance review, answer vendor questions, investigate incidents, train staff, and explain decisions to leaders or the public when appropriate.

Public Health Example

An agency using a generative AI assistant to draft public health messages should document the approved use, source materials, prohibited content, review process, accessibility expectations, language review, hallucination controls, user training, monitoring process, and incident response pathway. A system card would help reviewers understand the whole workflow rather than only the model.

Technical Deep Dive

AI documentation should be created before deployment and maintained after launch. The documentation should identify the system owner, intended users, data inputs, outputs, human review requirements, validation approach, known limitations, and change management process. A document that is created once and never updated will not support responsible operations.

Model cards are especially useful for predictive or classification models because they can document training data, evaluation metrics, subgroup performance, limitations, and intended use. System cards are especially useful for generative, RAG, and agentic systems because these tools depend on prompts, retrieval content, integrations, permissions, and human workflows.

Documentation should also cover changes. Model updates, prompt changes, data source changes, terminology updates, and workflow changes can all affect system behavior. Change logs and version histories help agencies reconstruct what happened if outputs change or incidents occur.

Risks, Failure Modes, and Guardrails

A common failure mode is treating the topic as a technical detail rather than a public health control. When staff do not connect technical decisions to authority, workflow, equity, privacy, security, and accountability, the agency may adopt a tool that appears useful but cannot be sustained or defended.

Another risk is relying on vendor claims without agency-defined evidence. Vendors may provide demonstrations, dashboards, or summary metrics that do not match the agency workflow or data environment. A guardrail is to require documentation, test results, acceptance criteria, and agency-controlled review.

A third risk is failing to plan for change. Public health data, policies, systems, and emergencies change. Any technical approach must include monitoring, review, versioning, escalation, and retirement pathways so the agency can respond when assumptions no longer hold.

Application to AI-Supported Workflows

Generative AI, predictive analytics, RAG, automation, and agentic workflows all depend on the controls described in this module. The controls should be scaled to the risk of the use case. A tool that drafts internal training text may require lighter review than a tool that updates case records, prioritizes outreach, or supports public communication during an emergency.

The module should be applied during readiness assessment, procurement, implementation planning, pilot testing, and post-deployment monitoring. Learners should document how the topic affects data, systems, users, safeguards, and decision-making for the selected workflow.

Reflection Questions

Where does this topic appear in one current or proposed AI workflow in your agency?

What decisions, data sources, users, or partners would be affected if this area is poorly designed?

What evidence would governance reviewers need before approving the workflow?

Which safeguards should be required before pilot testing?

Who owns ongoing monitoring and review?

Practical Exercise

Create a AI documentation packet for one real or realistic public health AI workflow. The artifact should describe the use case, affected users, data sources, system dependencies, risks, guardrails, review points, and unresolved questions.

The exercise should produce a work product that could be used in a governance meeting, procurement review, implementation planning session, pilot evaluation, or monitoring review. Learners should document assumptions and identify which staff or partners need to be consulted.

Expected Artifact or Evidence

The expected artifact is a completed AI documentation packet. It should be specific to a public health workflow and should include technical dependencies, governance questions, human review expectations, monitoring indicators, and unresolved risks.

Expected Assignment

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Knowledge Check

- 1. What is the best reason to address ai documentation, model cards, and system cards before deployment?
- 2. Which practice best supports responsible implementation?
- 3. Which statement best describes a common failure mode?
- 4. What should be included in the practical artifact for this module?
- 5. When should the issue be escalated for governance or leadership review?

References and Resources

- NIST AI Risk Management Framework: https://www.nist.gov/itl/ai-risk-management-framework
- NIST Generative AI Profile: https://www.nist.gov/publications/artificial-intelligence-risk-management-framework-generative-artificial-intelligence
- CDC Public Health Data Strategy and Data Modernization resources: https://www.cdc.gov/public-health-data-strategy/php/about/index.html
- OWASP Top 10 for LLM Applications, when security or AI integrations are relevant
- Agency privacy, cybersecurity, data governance, procurement, and records management policies